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{
"creator" : "ari.neko@cvut.cz",
"timestamp" : "09-07-2026_17:06:55",
"variant" : 3,
"subjectName" : "operating-systems",
"categories" : [ "File Systems", "Memory Management", "Kernel Architecture" ],
"questions" : [ {
"question" : "What is the primary role of the Virtual File System (VFS) layer in a kernel?",
"answers" : [ "To provide an abstract uniform interface for applications to access diverse file systems", "To manage virtual memory swap spaces on physical storage disks", "To buffer incoming network package flows before delivery to sockets", "To simulate internal hard drive arrays inside local ram spaces" ],
"correctIndex" : 0
}, {
"question" : "What is a major advantage of a journaling file system like ext4?",
"answers" : [ "It compresses text files to half their initial disk sector allocations", "It encrypts all user credentials automatically", "It records changes in a log to prevent corruption during sudden power losses", "It speeds up graphics card rendering rates" ],
"correctIndex" : 2
}, {
"question" : "What is the main difference between internal and external fragmentation?",
"answers" : [ "Internal is fixed by defragmentation; external requires virtual memory flushes", "External happens on hard drives; internal happens inside CPU registers", "External happens inside kernel space; internal happens in user space", "Internal occurs inside fixed allocated blocks; external occurs between variable blocks" ],
"correctIndex" : 3
}, {
"question" : "What is the role of a Translation Lookaside Buffer (TLB)?",
"answers" : [ "To act as a specialized cache for virtual-to-physical address translations", "To store execution metrics logs for background tasks", "To map file allocation records sequentially on disk", "To handle network data package buffering streams" ],
"correctIndex" : 0
}, {
"question" : "What state does a system enter when it spends more time swapping pages than executing instructions?",
"answers" : [ "Thrashing", "Starvation", "Paging", "Fragmenting" ],
"correctIndex" : 0
}, {
"question" : "What is a page fault?",
"answers" : [ "A physical hardware failure inside RAM modules", "An interrupt triggered when a program accesses a page not currently mapped in RAM", "A crash caused by syntax compile errors", "A data anomaly where two files share the same sector" ],
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"correctIndex" : 1
}, {
"question" : "What privilege layer does the operating system kernel run in on modern CPUs?",
"answers" : [ "Hypervisor mode", "User mode (Ring 3)", "Kernel mode (Ring 0)", "Safe mode" ],
"correctIndex" : 2
}, {
"question" : "What is a major advantage of a microkernel compared to a monolithic design?",
"answers" : [ "Smaller total storage size footprint", "Faster execution performance speeds", "Higher
system stability and isolation of corrupted drivers", "Lower context switching overhead costs" ],
"correctIndex" : 2
}, {
"question" : "What characterizes a monolithic kernel architecture?",
"answers" : [ "OS services are moved to separate user-space modules", "It runs exclusively on
virtual machines", "It eliminates context switching overhead entirely", "All operating system
services run inside a single privileged kernel space" ],
"correctIndex" : 3
}, {
"question" : "What happens during a hardware interrupt?",
"answers" : [ "The operating system completely reboots its kernel", "The CPU suspends its
execution stream to run an interrupt service routine", "Virtual memory maps flush physical hard
drive tracks", "The active process is terminated and marked as a zombie" ],
"correctIndex" : 1
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}
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